Applied Corporate Finance

Week #4C – Mergers and Acquisitions

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Plan of Attack

- **What M&A really is**
- **Value Creation and Division of Gains**
- **Sources of Synergies**
- **Why M&A fails**

Mergers and Acquisitions (M&A)

Let's clarify terminology:

- **Merger:** a new company is created by combining the merging companies, it does not involve cash payments, and there is a conversion ratio between the shares of the old companies and the new company resulting from the merger.

- **Acquisition:** one company acquires another; the acquirer must pay the shareholders of the acquired company in cash or shares, or both.

Reasons for Mergers and Acquisitions

- Are there synergies?
- Is the target company undervalued? What is the market missing?
- Market liquidity problem - is the stock mispriced (can work both ways!)?
- Strategic positioning - Are we defending or improving our business?
- Is the target company poorly managed? Can we manage it better?
- Is there any private interest from managers or some owners?

But in corporate finance, only one justification matters:

Does the deal increase the combined firm value?

Structure of Mergers and Acquisitions

Company A		+	Company B		=	CombinedCo	
A Assets	A Debt		B Assets	B Debt		A Assets	Old Debt (A+B)
	A Equity			B Equity		B Assets	New Debt
						CombinedCo Synergies	CombinedCo Equity

$$V_{A,B}(\text{after M\&}A) > V_A + V_B$$

Value created to acquirer's shareholders

- Company A acquires company B and pays $\boldsymbol{P_B}$ for 100% of company B.

- Gain for company B shareholders:

$$\textbf{Acquisition premium} = \boldsymbol{P_B} - \boldsymbol{V_B} \text{ (value of company B, before acquisition)}$$

- Gain for company A shareholders:

$$\textbf{Gain for company A shareholders} = \boldsymbol{V_{A,B}} - \boldsymbol{V_A} - \boldsymbol{P_B} = (\boldsymbol{V_{A,B}} - \boldsymbol{V_A} - \boldsymbol{V_B}) - (\boldsymbol{P_B} - \boldsymbol{V_B})$$

$$\textbf{Gain for company A shareholders} = \textbf{Synergies} - \textbf{Acquisition premium}$$

- **The value paid by company B - P_B - divides the value of synergies between the two parties.**

Division of Gains in Mergers

- In a merger, there is no cash payment. The equity of the new company is divided among the shareholders of the initial companies. Assume this example where each shareholder receives a share of the new company for each share of the old companies.

Company X

Company X Assets	5 000	Company X Debt	2 000
		Company X Equity	3 000
		# Shares	20
		Price per share	150

+

Company Y

Company Y Assets	15 000	Company Y Debt	12 000
		Company Y Equity	3 000
		# Shares	30
		Price per share	100

=

Company XY

Company X Assets	5 000	Company X + Y Debt	14 000
Company Y Assets	15 000	New Debt	-
Company XY Synergies	3 000	Company XY Equity	9 000
		# Shares	50
		Price per share	180

- The synergies of € 3,000 are distributed as follows: Company X shareholders gain € 600 (€30x20) and Company Y shareholders gain € 2,400 (€80x30).

Synergies

- **Operational Synergies:**
  - **Cost reductions (headcount, SG&A** (real estate, professional services, etc.)**)**
  - **Eliminate inefficiencies (overlapping structures and roles)**
  - **Economies of scale and/or utilization:** cost savings from increased production or utilization of installed capacity.

- **Revenues Synergies:**
  - **Cross-selling opportunities.**
  - **Market expansion**.
  - **increased market power** (though subject to antitrust scrutiny).
  - **New product development leveraging.**

Synergies (Cont.)

- **Financial Synergies:**
  - **Utilization of tax loss carryforwards.**
  - **Lower cost of capital:** reduced volatility of combined cash flows (diversification) can lead to a lower cost of debt or equity.
  - **Increase in borrowing capacity.**

- **Strategic Synergies:**
  - **Technology & R&D:** Acquiring critical patents, R&D capabilities, or specialized human capital.
  - **Vertical Integration:** Securing supply chains or distribution channels.
  - **Elimination of Inefficiencies:** Acquirer can run the target more effectively due to superior management or operational practices.

Synergies (Cont.)

$$\textbf{Synergies} = \frac{FCF_1}{1+r} + \frac{FCF_2}{(1+r)^2} + \cdots + \frac{FCF_T}{(1+r)^T}$$

- The value of merger or acquisition synergies can be determined through free cash flow discount methods.

Synergy Realization

- **Many acquisitions fail to create added value for the acquirer (it all goes to the shareholders of the acquired company)**

- **Synergies are easier to model than to execute!**

- The main reason stems from **failures in integrating the two companies after the acquisition:**
  - Human capital may leave if acquisitions are not handled carefully.
  - Poor assessment of cost reductions.
  - Failure to achieve expected "market power" due to "regulatory remedies" or competitor positioning.
  - Cultural problems.
  - Management distraction.
  - Execution problems.
  - Customer attrition.

Wrong Reasons for Mergers and Acquisitions

- **Earnings growth:**
  - If there are no synergies or other benefits, earnings growth can only be a fallacy and not true growth (accounting illusion).
- **Diversification:**
  - Shareholders can achieve their diversification at a probably lower cost!
- **Agency problems:**
  - Acquiring company managers may benefit from the merger or acquisition through increased compensation (salary, bonuses, and stock options) or greater security (larger companies are harder to be acquired).
  - A group of shareholders or managers may have conflict of interests.
  - Involved consultants and banks may have incentives to proceed with transactions due to the fees involved.

Takeaways

- **M&A creates value for all parties involved only if synergies exceed the premium paid.**
- **Valuation discipline matters more than strategic storytelling.**
- **Some M&A deals do not create value, they are made for wrong reasons!**
- **Execution risk is the main reason deals fail.**